Logic, Language, and Philosophy

Predicate Logic: semantics

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Set-theoretic basics

Sets

A set is a collection of objects. Examples of sets:

- the set of Chinese provinces
- the set of natural numbers
- the set consisting of Peter, Paul and Mary
- ...

Sets can be defined by:

- *enumeration*: $\{\text{Peter, Paul, Mary}\}$
- *condition*: $\{n \in \mathbb{N} \mid n \text{ is odd}\}$

Notation:

- $x \in A$, $x \notin A$: x is an element of A, x is not an element of A
- $\emptyset$: the empty set, i.e., the set without elements

Subset relation

- Subset:
  $A \subseteq B$ iff for all x: if $x \in A$ then $x \in B$
- Proper subset:
  $A \subset B$ iff for all x: if $x \in A$ then $x \in B$ and there is at least one x such that $x \in B$ and $x \notin A$

Extensionality

Extensionality Principle:
A set is completely defined by its elements

So: if $A \subseteq B$ and $B \subseteq A$, then $A = B$

Operations on sets

- Union: $A \cup B = \{x \mid x \in A \text{ or } x \in B\}$
- Intersection: $A \cap B = \{x \mid x \in A \text{ and } x \in B\}$
- Difference: $A - B = \{x \mid x \in A \text{ and } x \notin B\}$
- Powerset: $\mathcal{P}A = \{B \mid B \subseteq A\}$

Some simple facts:

a. If $a \in A$ and $A \subseteq B$, then $a \in B$

b. If $A \subseteq B$ and $B \subseteq C$, then $A \subseteq C$

c. $a \in A$ iff $\{a\} \subseteq A$

d. $A \cap B \subseteq A, \subseteq B$

e. $A, B \subseteq A \cup B$

f. $\emptyset \subseteq A$

g. $\{a\} \in \mathcal{P}A$ for all $a \in A$

Boolean operations and propositional connectives

Let W be a set of possibilities, and let $[\![\phi]\!] \subseteq W$ be the set of possibilities in which ϕ is true

Then we have:

- $[\![p]\!] \subseteq W$ for all proposition letters p
- $[\![\neg\phi]\!] = W - [\![\phi]\!]$
- $[\![\phi \vee \psi]\!] = [\![\phi]\!] \cup [\![\psi]\!]$
- $[\![\phi \wedge \psi]\!] = [\![\phi]\!] \cap [\![\psi]\!]$

Ordered pairs

A two-element set $\{x, y\}$ is the same set as $\{y, x\}$: only the elements define the set (extensionality), order plays no role.

An ordered pair $\langle a, b \rangle$ is a two-element construct in which order *does* matter: $\langle a, b \rangle \neq \langle b, a \rangle$ (unless $a = b$)

Definition (Cartesian product of A and B)

$A \times B = \{\langle a, b \rangle \mid a \in A \text{ and } b \in B\}$

Definition (A^2, Cartesian product of A and A)

$A \times A = \{\langle a, a' \rangle \mid a \in A \text{ and } a' \in A\}$

Ordered n-tuples

We also have ordered sequences of three, four, ... elements, i.e., n-tuples of elements from n sets:

Definition (Cartesian product of $A_1, A_2, \ldots A_n, \ldots$)

$$A_1 \times A_2 \cdots \times A_n \times \ldots =$$
$$\{\langle a_1, a_2, \ldots a_n, \ldots \rangle \mid a_1 \in A_1 \text{ and } a_2 \in A_2 \ldots \text{and } a_n \in A_n \ldots\}$$

Definition (A^n, n-fold Cartesian product of A)

$$A \times \cdots \times A = \{\langle a_1, \ldots, a_n \rangle \mid a_i \in A\}$$

Ordered n-tuples as sets

Ordered n-tuples can be defined in terms of sets.

Example:

Ordered pair $\langle a, b \rangle =_{\text{def}} \{a, \{a, b\}\}$

Note:

- $\{a, \{a, b\}\} \neq \{b, \{a, b\}\}$
- so, $\langle a, b \rangle \neq \langle b, a \rangle$

Relations, functions

Definition (n-place relation)

An n-place relation R in a set A is a subset of A^n

Definition (Function)

A function f from A to B is a subset F of $A \times B$ such that: if $\langle x, y\rangle \in F$ and $\langle x, z\rangle \in F$ then $y = z$

Notation:
$f : A \to B$, where A is called the *domain* of f and B its *range*.

Naive set theory and the Russell-paradox

Unlimited comprehension, i.e., definition of sets by *any* condition, allows:

- $V = \{x \mid x = x\}$, i.e., set containing everything

Note:

- $V \in V$

The Russell paradox:

- define: $R = \{x \mid x \notin x\}$
- question: $R \in R$?
- answer:
  - if $R \in R$, then $R \notin R$
  - if $R \notin R$, then $R \in R$
- contradiction

Solutions

Solutions:

- type theory (Russell)
- limited comprehension (Zermelo-Fraenkel, ZFC)
- distinction sets – classes (von Neumann-Bernays-Gödel)
- ...

Syntax: recap

Vocabulary

Definition (Vocabulary)

The vocabulary of a language of predicate logic L consists of:

1. A set of predicate letters $A_1^i, \ldots, A_n^i, \ldots, A_1^k, \ldots, A_n^k, \ldots$ of arbitrary -arity $i, k, \ldots$; indicated as PRED_L
2. A set of individual constants $c_1, \ldots, c_n, \ldots$; indicated as CON_L
3. A set of variables $x_1, \ldots, x_n, \ldots$; indicated as VAR_L

Definition (Terms)

The terms of L consists of the individual constants and the variables of L; indicated as TERM_L

Notation

Note that for mnemonic purposes we will also use:

- $P, Q, \ldots R, S, \ldots$, as symbols for predicate letters
- $a, b, \ldots, p, s, \ldots$ as symbols for individual constants
- $x, y, z, \ldots v, w, \ldots$ as symbols for variables

And we leave out the indication of the -arity of a predicate letter when this is clear from the context.

Formulas

Definition (Formulas)

Given a vocabulary for L, the set of formulas of a language of predicate logic L is the smallest set such that:

1. If A_i^n is an n-ary predicate letter and $t_1, \ldots, t_n$ are terms, then $A_i^n t_1 \ldots t_n$ is a formula.
2. If ϕ is a formula, then so is $\neg\phi$.
3. If ϕ, ψ are formulas, then so are $\phi \wedge \psi, \phi \vee \psi, \phi \rightarrow \psi, \phi \leftrightarrow \psi$.
4. If ϕ is a formula and x_j is a variable, then $\forall x_j \phi, \exists x_j \phi$ are formulas.

Stipulation: We leave off the outer brackets.

Quantifiers and scope

Definition (Scope of a quantifier occurrence)

If $\forall x \psi$ is a sub-formula of ϕ, then ψ is called the *scope* of the occurrence of the quantifier $\forall x$ in ϕ.

Free and bound variable occurrences

Definition (Free and bound occurrences of variables)

An occurrence of a variable x in a formula ϕ which is not part of a quantifier is said to be *free* in ϕ if this occurrence of x is not in the scope of the occurrence of a quantifier $\forall x$ ($\exists x$) in ϕ

If $\forall x\psi(\exists x\psi)$ is a sub-formula of ϕ and an occurrence of x is free in ψ, then the occurrence of x is said to be *bound* by that occurrence of the quantifier $\forall x\psi$ ($\exists x\psi$).

Formulas and sentences

Definition (Sentence)

A sentence is a formula of L that lacks occurrences of free variables

Semantics: introduction

Extending referential semantics

Proposition logic:

- truth values
- truth functions

Predicate logic:

- truth values of atomic sentences of the form $A^i c_1 \ldots c_i$ depend on what A^i and $c_1 \ldots c_i$ *refer to*
- truth values of quantified sentences $\forall x \phi$, $\exists x \phi$ depend on which elements in the domain *satisfy* ϕ

Example

Domain: people; Bx: x is bald; Lxy: x loves y; b: Bill; j: John; m: Mary.

1. Bb is true if the property that is referred to by B applies to the individual that is referred to by b, i.e., iff Bill is bald
2. Lmj is true iff the relation that is referred to by L holds between the individuals referred to by m and j, respectively, i.e., iff Mary loves John
3. $\forall x \exists y Lxy$ is true iff for every individual in the domain there is an individual in the domain such that the relation that is referred to by L holds between the first one and the second one, i.e., iff everyone loves someone

Semantics: ingredients

Models

Definition (Model)

A model M for a predicate logical language L is an ordered pair $\langle D_M, I_M \rangle$ such that:

- D_M is a non-empty set
- I_M is a function such that:
  - $I_M(A_i^n) \subseteq D_M^n$, for all predicate letters A_i^n of L
  - $I_M(c_i) \in D_M$, for all individual constants c_i of L

D_M is the domain of discourse. It can be any set as long as it is not empty, and does not contain everything.

I_M is the interpretation function. It specifies what the predicate letters and individual constants refer to.

Truth definition: first step

Definition (Valuations)

Let M be a model $\langle D_M, I_M \rangle$ for L. A valuation V_M is a function with the sentences in L as its domain and $\{0, 1\}$ as its range, such that:

1. $V_M(A_i^n c_1 \dots c_n) = 1$ iff $\langle I_M(c_1) \dots I_M(c_n) \rangle \in I_M(A_i^n)$
2. $V_M(\neg\phi) = 1$ iff $V_M(\phi) = 0$
3. $V_M(\phi \wedge \psi) = 1$ iff $V_M(\phi) = 1$ and $V_M(\psi) = 1$
4. as usual for $\vee, \rightarrow, \leftrightarrow$

Atomic sentences

Example:

- $V_M(Bb) = 1$ iff the entity $I_M(b)$ is an element of the set of entities $I_M(B)$
- $V_M(Ljm) = 1$ iff the ordered pair $\langle I_M(j), I_M(m)\rangle$ is an element of the set of ordered pairs $I_M(L)$

Model checking

- Domain: $\{\mathtt{john}, \mathtt{bill}, \mathtt{mary}\}$
- $I_M(j) = \mathtt{john}, I_M(b) = \mathtt{bill}, I_M(m) = \mathtt{mary}$
  - Let M be such that $I_M(B) = \{\mathtt{john}, \mathtt{bill}\}$.
    Then $V_M(Bb) = 1$ since $I_M(b) = \mathtt{bill}$ and $\mathtt{bill} \in I_M(B)$
  - Let M be such that $I_M(L) = \emptyset$.
    Then $V_M(Ljm) = 0$, since $\langle I_M(j), I_M(m)\rangle = \langle \mathtt{john}, \mathtt{mary}\rangle \notin \emptyset$
  - Let M be such that $I_M(L) = \{\langle \mathtt{john}, \mathtt{mary}\rangle, \langle \mathtt{mary}, \mathtt{bill}\rangle\}$.
    Then $V_M(Ljm) = 1$ since $\langle I_M(j), I_M(m)\rangle = \langle \mathtt{john}, \mathtt{mary}\rangle \in I(L)$

Quantifiers

Still to be done: $\forall x \phi$ and $\exists x \phi$

Two options:

- Substitutional interpretation
- Interpretation via assignments

Substitutional interpretation

Basic idea

Observation:

- universal quantifier is connected with conjunction:
  $\forall x Px \approx Pc_1$ and Pc_2 and ... and Pc_n and ...
- existential quantifier is connected with disjunction:
  $\exists x Px \approx Pc_1$ or Pc_2 or ... or Pc_n or ...

Suggestion: if we can express all the instances we can explain quantifiers in terms of (ultimately) atomic sentences.

For this to work *everything in the domain has to have a name*.

Additional requirement

Add to the definition of a model the requirement that:

- I_M is a function of CON_L *onto* D_M

Definition

f is a function from A *onto* B iff for all $b \in B$ there is at least one $a \in A$ such that $f(a) = b$ ('No b's left behind')

Consequence

Note:

- If D_M is infinite, then so is CON_L

The assumption of an infinite vocabulary is not a problem for a logical language, but is unrealistic for a natural language.

Substitution

Notation: $[c/x]\phi$ is the formula ϕ' that results from substituting c for every free occurrence of x in ϕ.

Examples:

- $[c/x]Px = Pc$
- $[c/x](Px \wedge Qx) = Pc \wedge Qc$
- $[c/x](Px \rightarrow \forall xQx) = Pc \rightarrow \forall xQx$

Note:

- $[c/x]([d/y]\phi)$: substitutions can be “stacked”

Truth definition

Definition (Valuations)

Let M be a model $\langle D, I \rangle$ for L. A valuation V_M is a function with the sentences in L as its domain and $\{0, 1\}$ as its range, such that:

1. $V_M(A_i^n c_1 \dots c_n) = 1$ iff $\langle I_M(c_1) \dots I_M(c_n) \rangle \in I_M(A_i^n)$
2. $V_M(\neg\phi) = 1$ iff $V_M(\phi) = 0$
3. $V_M(\phi \wedge \psi) = 1$ iff $V_M(\phi) = 1$ and $V_M(\psi) = 1$
4. as usual for $\vee, \rightarrow, \leftrightarrow$
5. $V_M(\forall x \phi) = 1$ iff $V([c/x]\phi) = 1$ for all c in CON_L
6. $V_M(\exists x \phi) = 1$ iff $V([c/x]\phi) = 1$ for at least one c in CON_L

Examples

1. $V_M(\exists x Bx) = 1$
   iff there is a c: $V_M([c/x]Bx) = 1$,
   iff there is a c: $V_M(Bc) = 1$
   iff there is a c: $I_M(c) \in I_M(B)$
2. $V_M(\forall x \exists y Lxy) = 1$
   iff for all c: $V_M([c/x]\exists y Lxy) = 1$,
   iff for all c: $V_M(\exists y Lcy) = 1$,
   iff for all c there is a c': $V_M([c'/y]Lcy) = 1$,
   iff for all c there is a c': $V_M(Lcc') = 1$
   iff for all c there is a c': $\langle I_M(c), I_M(c') \rangle \in I_M(L)$

Model checking

Let M be the following model:

- $D_M = \{\texttt{john}, \texttt{bill}, \texttt{mary}\}$
- $I_M(j) = \texttt{john}$, $I_M(b) = \texttt{bill}$, $I_M(m) = \texttt{mary}$
- $I_M(B) = \{\texttt{john}, \texttt{bill}\}$
- $I_M(L) = \{\langle \texttt{john}, \texttt{mary}\rangle, \langle \texttt{mary}, \texttt{bill}\rangle\}$

Check $\exists x(Bx \wedge \exists y Lyx)$:

- $V_M(\exists x(Bx \wedge \exists y Lyx)) = 1$, since
- $V_M([b/x](Bx \wedge \exists y Lyx)) = V_M(Bb \wedge \exists y Lyb) = 1$, since
  - $V_M(Bb) = 1$, since
    - $I_M(b) = \texttt{bill}$ and $\texttt{bill} \in I_M(B)$
  - and
  - $V_M([m/y]Lyb) = V_M(Lmb) = 1$, since
    - $I_M(m) = \texttt{mary}$ and $\langle \texttt{mary}, \texttt{bill}\rangle \in I_M(L)$

Observations

Note:

- Awkward assumption that the language has a name for all elements in the domain
- Truth definition only interprets *sentences*:
  - no interpretation supplied for atomic formulas with variables
  - quantified formulas $\forall x\phi$, $\exists x\phi$, are not interpreted in terms of ϕ, but in terms of $[c/x]\phi$ (breach of compositionality)

Interpretation with assignments

Assignments

Definition (Assignments)

Given a model M, the set of assignments G is the set of all functions g from VAR_L to D_M

Notation:

- $g[x/d]$ is g except for the possible difference that $g[x/d](x) = d$
- general case: $g[x_i/d_k, \ldots, x_n/d_l]$

Interpretation of terms

Definition (Interpretation of terms)

$$[\![t]\!]_{M,g} = \begin{cases} I_M(t) & \text{if } t \text{ is a constant} \\ g(t) & \text{if } t \text{ is a variable} \end{cases}$$

Truth definition

Definition (Valuations)

Let M be a model $\langle D, I \rangle$ for L and G the set of assignments. A valuation $V_{M,g}$ is a function from the set of formulas of L to $\{0, 1\}$ such that:

1. $V_{M,g}(At_1 \dots t_n) = 1$ iff $\langle [\![t_1]\!]_{M,g}, \dots, [\![t_n]\!]_{M,g} \rangle \in I(A)$
2. $V_{M,g}(\neg\phi) = 1$ iff $V_{M,g}(\phi) = 0$
3. $V_{M,g}(\phi \wedge \psi) = 1$ iff $V_{M,g}(\phi) = 1$ and $V_{M,g}(\psi) = 1$
4. as usual for $\vee, \rightarrow, \leftrightarrow$
5. $V_{M,g}(\exists x\phi) = 1$ iff there is a $d \in D$: $V_{M,g[x/d]}(\phi) = 1$
6. $V_{M,g}(\forall x\phi) = 1$ iff for all $d \in D$: $V_{M,g[x/d]}(\phi) = 1$

Example

$V_{M,g}(\exists x(Px \wedge Qx)) = 1$
iff there is a $d \in D$: $V_{M,g[x/d)]}(Px \wedge Qx) = 1$
iff there is a $d \in D$: $V_{M,g[x/d)]}(Px) = 1$ and $V_{M,g[x/d)]}(Qx) = 1$
iff there is a $d \in D$: $[\![x]\!]_{M,g[x/d)]} \in I_M(P)$ and $[\![x]\!]_{M,g[x/d)]} \in I_M(Q)$
iff there is a $d \in D$: $g[x/d](x) \in I_M(P)$ and $g[x/d](x) \in I_M(Q)$
iff there is a $d \in D$: $d \in I_M(P)$ and $d \in I_M(Q)$

Model checking

Let M be as before.

Check $\exists x(Bx \wedge \exists y Lxy)$:

- $V_{M,g}(\exists x(Bx \wedge \exists y Lxy)) = 1$, since
- $V_{M,g[x/\mathtt{j}]}(Bx \wedge \exists y Lxy) = 1$, since
  - $V_{M,g[x/\mathtt{j}]}(Bx) = 1$, since
    - $[\![x]\!]_{M,g[x/\mathtt{j}]} = g[x/\mathtt{j}](x) = \mathtt{j}$ and $\mathtt{j} \in I_M(B)$
  - and:
  - $V_{M,g[x/\mathtt{j},y/\mathtt{m}]} Lxy = 1$, since
    - $\langle [\![x]\!]_{g[x/\mathtt{j},y/\mathtt{m}]}, [\![y]\!]_{g[x/\mathtt{j},y/\mathtt{m}]} \rangle = \langle g[x/\mathtt{j},y/\mathtt{m}](x), g[x/\mathtt{j},y/\mathtt{m}](y) \rangle = \langle \mathtt{j},\mathtt{m} \rangle$, and
    - $\langle \mathtt{j},\mathtt{m} \rangle \in I_M(L)$

Truth and validity

Truth

Definition (Truth, substitution)

A sentence ϕ is true w.r.t. M iff $V_M(\phi) = 1$

Definition (Truth, assignments)

A formula ϕ is true w.r.t. M and g iff $V_{M,g}(\phi) = 1$

Validity

Definition (Validity, substitution)

A sentence ϕ is valid iff $V_M(\phi) = 1$, for all M

Definition (Validity, assignments)

A formula ϕ is valid iff $V_{M,g}(\phi) = 1$, for all M and g

Sentences, formulas

Note:

- Substitutional interpretation: truth and validity are restricted to sentences
- Assignments: truth and validity cover all formulas

Question: is there “real life” reasoning with formulas containing free variables?

Perhaps: demonstratives, ‘free’ pronouns

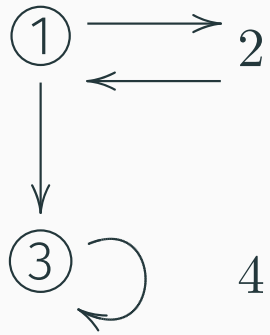
Model checking

Model representations

Let M be the following model:

- $D_M = \{1, 2, 3, 4\}$
- $I_M(P) = \{1, 3\}$
- $I_M(R) = \{\langle 1, 2\rangle, \langle 2, 1\rangle, \langle 1, 3\rangle, \langle 3, 3\rangle\}$

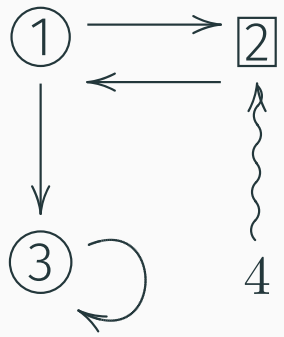
Then M is represented by the following graph:



Objects are represented by points, the arrows indicate the R-relation and the property P is represented by points being encircled.

Enriching graphs

If need be, we enrich the graph with other elements to represent more properties and relations:



Exercise: example

Consider the following model for a predicate logical language with a 1-place predicate letter A, and 2-place predicate letter R:

$$D = \{1, 2, 3, 4\}$$
$$I(A) = \{1, 4\}$$
$$I(R) = \{\langle 1, 1\rangle, \langle 1, 2\rangle, \langle 1, 3\rangle, \langle 1, 4\rangle, \langle 3, 3\rangle, \langle 4, 1\rangle, \langle 4, 2\rangle, \langle 4, 3\rangle, \langle 4, 4\rangle\}$$

a. Draw a graph of the model
b. Which of the following formulas are true in the model? Motivate your answer.
    i. $\forall x(Ax \leftrightarrow \forall y Rxy)$
    ii. $\forall x \forall y(Rxy \rightarrow Ryx)$
    iii. $\exists x \exists y(Rxy \wedge Ryx)$

Homework

Homework

- Read chapter 13 and make exercises 1,2
- Read chapter 14 and Tarski's 'The Semantic Conception of Truth'